

ACM Summer School on Symmetric Key Cryptography

Introduction to Cryptography: Part 1

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- 1 Classical Cryptography
 - Shift Cipher
 - Monoalphabetic Substitution
 - Vigenere Cipher
 - Cryptanalysis
- 2 Kerckhoffs' Principle



The Basic Communication Problem

Alice $\longrightarrow$ Bob

while Eve listens.

Questions:

- Can Alice send confidential information?
- Can Bob verify authenticity?
- Can Eve learn anything?

Can also be asked for data at rest.



Encryption Schemes

A *symmetric* encryption scheme consists of three algorithms

$$\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$$

- $\text{Gen}(1^\lambda)$ generates a secret key k
- $\text{Enc}_k(m)$ outputs ciphertext c
- $\text{Dec}_k(c)$ outputs plaintext m

Correctness:

$$\Pr[\text{Dec}_k(\text{Enc}_k(m)) = m] = 1$$

for all valid messages.



Ancient History

- Non-standard hieroglyphics in ancient Egypt (circa 2nd millenium BCE)
- Scytale cipher (ancient Greece/Sparta)



Figure: The Scytale. Img src: Wikipedia

- ... and many others!



Classical Cryptography

Before computers, cryptography relied mainly on:

- 1 Substitution
- 2 Transposition
- 3 Secret procedures

Examples:

- Caesar cipher
- Monoalphabetic substitution
- Vigenère cipher
- Enigma machine



Shift Cipher

Also known as Caesar Cipher.

Alphabet:

$$A = 0, B = 1, \dots, Z = 25$$

Encryption:

$$c = (m + k) \bmod 26$$

Decryption:

$$m = (c - k) \bmod 26$$



Shift Cipher Example

Key:

$$k = 3$$

Encryption:

HELLO

$$H \rightarrow K$$

$$E \rightarrow H$$

$$L \rightarrow O$$

Result:

HELLO → KHOOR



Cryptanalysis of Shift Cipher

Key space:

$$|\mathcal{K}| = 26$$

Attack:

- Try all possible keys.
- Check which plaintext makes sense.

Complexity:

$$O(26)$$

which is trivial.



Monoalphabetic Substitution Cipher

Each letter is replaced by another letter.

Example:

$$A \rightarrow Q$$

$$B \rightarrow W$$

$$C \rightarrow E$$

etc.

Key space:

$$26!$$

$$\approx 4 \times 10^{26}$$



Why Large Key Space Is Not Enough

Despite huge key space:

26!

monoalphabetic substitution is insecure.

Reason:

Encryption preserves statistical structure of language.

Example:

E, T, A, O, I, N

remain the most frequent symbols.



Frequency Analysis

Letter frequencies in English:

Letter	Frequency
E	12.7%
T	9.1%
A	8.2%
O	7.5%
I	7.0%
N	6.7%

Frequency analysis often recovers the entire key.



Polyalphabetic Encryption

Idea:

Use multiple substitution alphabets.

Most famous example:

Vigenere Cipher

A key word determines which shift is applied.



Vigenère Cipher

Let

$$k_i$$

denote the key symbol.

Encryption:

$$c_i = (m_i + k_i) \bmod 26$$

Decryption:

$$m_i = (c_i - k_i) \bmod 26$$

Different positions use different shifts.



Example

Message:

ATTACKATDAWN

Key:

ACMSUMMERSCH

Encryption performed symbol-by-symbol.

Ciphertext:

AVFSWWMXUSYU

Resulting ciphertext appears significantly more random than a shift cipher.



Breaking the Vigenère Cipher

Main observation:

- Key repeats periodically.

Goal:

- 1 Determine key length n .
- 2 Then it becomes easy: Recover each shift independently.
- 3 It can therefore be viewed as n interleaved Caesar ciphers



Kasiski Examination

Invented by Friedrich Kasiski (1863).

Procedure:

- 1 Locate repeated ciphertext patterns. Plaintext:
SUMMERSCHOOLISCOOLINSUMMERBUTMONSOONISCOOLER
key: ABCD
Ciphertext: SVOPE SUFHPQOITEROMKQSVOPESDXTNQQSPQQITEROMGU
- 2 Compute distances between occurrences.
- 3 All factors of a distance are potential key lengths.
- 4 Here the distance between "SVOPE" is 20
- 5 Potential key lengths: 20, 10, 5, 4, 2, 1.
- 6 Distance between TEROM is 24
- 7 Potential key lengths: 24, 12, 8, 6, 4, 3, 2, 1
- 8 Take greatest common divisors: 4.
- 9 Infer likely key length.



Friedman technique

Again, consider **Plaintext**:

SUMMERSCHOOLISCOOL

Key:

ABCD

Using shifts

$$A = 0, \quad B = 1, \quad C = 2, \quad D = 3$$

the ciphertext becomes

SVOPESTFKROMIVEOPM

The key length is $k = 4$



Streams Produced by the Key

SVOPESTFKROMIVEOPM

Pos	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Cipher	S	V	O	P	E	S	T	F	K	R	O	M	I	V	E	O	P	M

Grouping positions according to the key:

Stream 1 : *S, E, K, I, P*

Stream 2 : *V, S, R, V, M*

Stream 3 : *O, T, O, E*

Stream 4 : *P, F, M, O*

Each stream is encrypted with a single Caesar shift.



Index of Coincidence

The Index of Coincidence (IC) is the probability that two randomly chosen letters are identical.

For a text of length N ,

$$IC = \frac{\sum_i f_i(f_i - 1)}{N(N - 1)}$$

where f_i is the frequency of letter i .

For the ciphertext

SVOPESTFKROMIVEOPM

the frequencies are

$E = 2, F = 1, I = 1, K = 1, M = 2, O = 3, P = 2, R = 1, S = 2, T = 1$ and $V = 2$



Thus

$$\sum_i f_i(f_i - 1) = 2 + 2 + 6 + 2 + 2 + 2 = 16$$

and

$$IC = \frac{16}{18 \cdot 17} = 0.0523.$$



Why Does IC Depend on the Key Length?

Let

$$I_P \approx 0.065$$

be the IC of English text.

Let

$$I_R \approx \frac{1}{26} = 0.0385$$

be the IC of random text.

Choose two random ciphertext positions.

- Same stream: $P(\text{same stream}) = \frac{1}{k}$
Coincidence probability: I_P
- Different streams: $P(\text{different streams}) = 1 - \frac{1}{k}$
Coincidence probability: I_R



Derivation of the Friedman Estimate

Expected Index of Coincidence

$$E[IC] = I_C = \frac{1}{k} I_P + \left(1 - \frac{1}{k}\right) I_R$$

Substituting

$$I_P = 0.065, \quad I_R = 0.0385$$

gives

$$I_C = 0.0385 + \frac{0.0265}{k}.$$

Solving for k ,

$$I_C - 0.0385 = \frac{0.0265}{k}$$



Estimating the Key Length

From the ciphertext we computed

$$I_C = 0.0523.$$

Applying the Friedman estimate,

$$k = \frac{0.0265}{0.0523 - 0.0385}$$

$$k = 1.92.$$

The estimate is not exact because the ciphertext is very short ($N = 18$).

For longer ciphertexts, the estimate converges toward the true key length.

Actual key length: $k = 4$



Index of Coincidence (The Friedman test)

Another technique for key-length estimation.

Defined as

$$IC = \frac{\sum_i f_i(f_i - 1)}{N(N - 1)}$$

where f_i is frequency of symbol i .

English text and random text exhibit different IC values.



Lessons from Classical Cryptography

Classical systems failed because:

- Statistical structure leaked information.
- Security definitions were informal.
- Small effective key spaces.
- Human operation errors.

These lessons motivated modern cryptography.



The Enigma Machine

Used extensively by Germany during World War II.

Components:

- Keyboard
- Lampboard
- Plugboard
- Rotors
- Reflector

Each keypress altered the encryption mapping.



Rotor-Based Encryption

Rotors implement permutations

$$\pi_1, \pi_2, \pi_3$$

of the alphabet.

After each character:

- Rotor positions change.
- Effective substitution changes.
- Cipher becomes polyalphabetic.



Why Enigma Appeared Secure

Advantages:

- Huge number of configurations.
- Dynamic substitutions.
- Frequent key changes.

Estimated key space:

$$> 10^{20}$$

which was enormous for its time.



Breaking Enigma

Major contributors:

- Marian Rejewski
- Jerzy Różycki
- Henryk Zygalski
- Alan Turing
- Gordon Welchman

Techniques used:

- Mathematical analysis
- Known plaintext attacks
- Operational mistakes



Lessons from Enigma

Security failures arose from:

- Human errors
- Repeated message formats
- Protocol weaknesses
- Operational practices

Important lesson:

Strong algorithms alone do not guarantee security.



Security Goals

A secure encryption scheme should provide:

Confidentiality

Adversary learns no useful information about plaintext.

Integrity

Unauthorized modifications are detectable.

Authentication

Identity of communicating parties can be verified.



Kerckhoffs' Principle

Auguste Kerckhoffs (1883):

A cryptosystem should remain secure even if everything about the system except the key is public knowledge.

Implications:

- Security must rely only on secrecy of keys.
- Algorithms should be publicly scrutinized.
- Avoid “security through obscurity”.



Why Kerckhoffs Was Right

History repeatedly demonstrates:

- Algorithms eventually leak.
- Hardware can be reverse engineered.
- Employees can disclose secrets.
- Attackers can inspect software.

Modern cryptography assumes the adversary knows the complete specification of the system.



Thank You

Questions?

